

# AI Analysis

## Al Safa 2 Park — AI Methodology Log for Phase 1

### 1. Purpose

This log documents how AI was used to assist Phase 1 (Existing Site Understanding), satisfying the Brief’s explicit requirement (Schedule 1, Section C) that participants document AI usage throughout the design process — starting from this first phase.

### 2. Tools Used in Phase 1

| Tool                                  | Role                                                                                                                                                                 |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claude (Anthropic)                    | Orchestration of the analysis workflow; synthesis of written findings from source PDFs/images; SWOT synthesis logic; identification of data gaps vs. available facts |
| Python (pandas, matplotlib, numpy)    | Quantitative computation and chart generation                                                                                                                        |
| pvlib (NREL Solar Position Algorithm) | Exact astronomical solar position computation for the site’s precise lat/lon — Climate Analysis (1.05) and Shadow Analysis (1.06)                                    |
| Pillow / numpy (image processing)     | Color-threshold pixel analysis of the master-plan graphic to estimate visible green-canopy footprint and site bounding box                                           |
| ReportLab                             | Programmatic PDF report generation with embedded charts, tables, and source-code proof appendices                                                                    |

### 3. Areas of Application

- **Site data analysis and interpretation** — automated extraction from the master-plan image and structured compilation of brief/manual text.
- **Mapping and visualization of spatial information** — sun-path polar diagram, shadow direction plan-view diagram, wind rose, SWOT matrix — all generated programmatically, ensuring traceable, reproducible numbers.
- **Environmental analysis** — solar radiation, temperature, humidity, wind, and shadow-casting computed using real astronomical formulas, not estimated by eye.

### 4. Role of Human Judgment (this session)

- AI (Claude) was directed by the user’s explicit phase structure and file scope — it did not invent the 10-phase framework or analysis categories.
- Every claim in the Phase 1 documents is tagged as either (a) sourced directly from the competition brief/manual/image, (b) computed via a verifiable formula/library, or (c) an explicitly flagged data gap.
- No design decisions (zoning, layout, planting choices, etc.) were made in this phase.

### 5. Reproducibility

All charts/tables in Phase 1 were generated by scripts in 01\_PHASE1\_EXISTING\_PARK/\_scripts/ and can be re-run to regenerate identical outputs — satisfying the Brief’s ask for an AI methodology and workflow that shows genuine, traceable process rather than a black box.